CODEBASE-AUDITED TECHNICAL REFERENCE

Weaver Quantum Equations & Notation

Every identifiable quantum, quantum-inspired, and supporting mathematical expression used by the runtime or present in Weaver's declared training materials.

12
measured circuit qubits in the active runtime

144
sparse reservoir addresses, not simulated qubits

156
total represented architecture roles

Weaver v3
Repository evidence edition

Audited 25 July 2026
Runtime • fallback • datasets • provenance

READ THIS FIRST

Scope, certainty, and status labels

This reference reports what is demonstrable from the checked-out Weaver codebase. It separates executable mathematics from architectural language and from equations that appear only in generated training text.

Most important finding. Weaver's live quantum path builds and measures a 12-qubit Qiskit circuit. The other 144 "reservoir qubits" are sparse software addresses populated from 12 measured marginals. The live state space is therefore $\mathbb{C}^{2^{12}}$, not a dense $\mathbb{C}^{2^{156}}$ state vector.

STATUS	MEANING IN THIS DOCUMENT	STRENGTH OF CLAIM
LIVE RUNTIME	Executed by the active 12-qubit orchestrator.	Directly verified in source and focused unit tests.
LEGACY FALLBACK	Executable 7-qubit circuit used when the active network cannot initialize.	Directly verified in source and circuit construction.
CLASSICAL SUPPORT	Classical gating, ODE, scoring, or geometric math around the quantum path.	Directly verified in executable source.
DORMANT	Implemented equation-bearing code, but no active runtime import was found.	Present in source; not established as wired into production flow.
TRAINING TEXT	Equation appears in a declared fine-tuning dataset or its generator.	Corpus presence proven; completed ingestion into every model is not.
EVIDENCE ONLY	Equation reconstructed from a separate sampled pretraining checkpoint corpus.	Checkpoint exposure evidence, not active Weaver runtime behavior.

"Was trained on" has a hard boundary. The repository proves that several quantum datasets were declared as inputs to training scripts. It does not contain complete provenance for the upstream Qwen2 3B base model, and the declared Nemotron output directory was not present. Accordingly, this is the complete *codebase-observable* equation set, not a claim about every token ever seen by an upstream foundation model.

Contents

01 · NOTATION DICTIONARY

Symbols used throughout Weaver

The same Greek letter is used for different ideas in different files. This glossary disambiguates them.

SYMBOL	DEFINITION	UNITS / DOMAIN	USED IN
q, i, j	Core-qubit or pathway index.	Integer index	Circuits, marginals, geometry
ℓ	Variational layer index.	$\ell = 0,\ldots,L-1$	Active circuit
N_c	Measured core width.	12 qubits	Active runtime
N_r	Sparse reservoir-address count.	144 addresses	Reservoir projection
N_K	Represented Kingston architecture width.	$N_c+N_r=156$	Metadata / diagrams
φ_g	Golden ratio used for dodecahedral coordinates.	$(1+\sqrt{5})/2 \approx 1.618$	12-node graph geometry
φ_p	Pentagonal phase step.	$2\pi/5 = 72° \approx 1.2566$ rad	Fallback circuit, Pineal geometry, training text
θ, ϕ	Trainable Y- and Z-rotation angles.	Radians, wrapped mod 2π	Active variational circuit
$\alpha_{\ell i}$	Deterministic layer/qubit state-encoding phase.	Radians	Active circuit
$\beta_{\ell ab}$	Pair phase passed to controlled-RZ.	Radians	Active circuit
$\gamma_{\ell i}$	Open-system proxy RZ drive angle.	Radians	Active circuit
c_s	Observed shot count for bit string s.	Non-negative integer	Qiskit measurement counts
p_s	Empirical probability c_s/N.	[0,1]	Shannon entropy, marginals
H	Context-dependent entropy. In runtime it is Shannon entropy; in training prose it may mean von Neumann entropy.	Bits if $\log_2$; nats if ln	Runtime and datasets
ρ	Density operator appearing in training text, not the live circuit implementation.	Positive trace-one operator	Omega dataset
Γ	Decoherence rate in generated Lindblad examples.	Inverse time	Omega dataset

SYMBOL	DEFINITION	UNITS / DOMAIN	USED IN
L	Lindblad jump operator in generated training text; also layer count elsewhere.	Operator / integer	Context dependent
τ	Adaptive governor threshold or liquid-fracture time constant, depending on file.	Scalar / time	Governor, liquid fracture
Δt	Elapsed time or ODE integration step.	Seconds	Governor, liquid fracture

> **Do not conflate φ_g and φ_p.** The code calls both values “phi” in different modules. One is the golden ratio; the other is a 72-degree phase increment. They are numerically and conceptually different.

What “156 qubits” means in the implementation

LIVE RUNTIME 12 measured qubits + sparse reservoir metadata

REPRESENTED ARCHITECTURE COUNT (2.1)

$$N_K = N_c + N_r = 12 + 144 = 156$$

`quantum_networks.py: N_CORE_QUBITS, N_RESERVOIR_QUBITS, N_KINGSTON_QUBITS`

ACTUALLY SIMULATED HILBERT SPACE (2.2)

$$\mathscr{H}_{\text{live}} = \mathbb{C}^{2^{12}}, \quad \dim(\mathscr{H}_{\text{live}}) = 4096$$

The active Qiskit circuit is constructed with 12 quantum bits and 12 classical readout bits.

SPARSE RESERVOIR ADDRESS SET (2.3)

$$\mathscr{R} = \{12, 13, \ldots, 155\}, \quad |\mathscr{R}| = 144$$

Each reservoir entry is a derived record containing source core, band, weight, decay, and phase address. It is not an amplitude in a 156-qubit state vector.

Not implemented: $\mathcal{H} = \mathbb{C}^{2^{156}}$, a 156-qubit state vector, a 156-qubit density matrix, or a partial trace from qubits 12–155 during live inference.

Sparse connectivity bookkeeping

CORE-TO-RESERVOIR ADDRESS MAPPING (2.4)

$$r(q, b) = 12 + [qN_c + b] \bmod N_r$$

For $q, b \in \{0, \ldots, 11\}$, every core role owns 12 reservoir addresses.

`quantum_networks.py: core_reservoir_couplings()`

Dodecahedral coordinates and pentagonal phase

LIVE RUNTIME

Golden-ratio geometry

GOLDEN RATIO (3.1)

$$\varphi_g = \frac{1+\sqrt{5}}{2}$$

The 12 raw vertices are permutations/sign patterns of:

$$(\pm 1, \pm\varphi_g, 0), (0, \pm 1, \pm\varphi_g), (\pm\varphi_g, 0, \pm 1)$$

COORDINATE NORMALIZATION (3.2)

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

SQUARED EUCLIDEAN EDGE METRIC (3.3)

$$d_{ij}^2 = \sum_{k=1}^{3} (v_{ik} - v_{jk})^2$$

The five nearest neighbors per vertex produce 30 unique edges and degree 5 at every node.

FALLBACK + SUPPORT

Pentagonal phase geometry

PHASE STEP (3.4)

$$\varphi_p = \frac{2\pi}{5} = 72° \approx 1.256637$$

PINEAL VERTEX ANGLE AND POSITION (3.5)

$$\vartheta_i = \frac{2\pi i}{5}, \quad v_i = (\cos\vartheta_i, \sin\vartheta_i)$$

This constant drives the 7-qubit fallback rotations, the classical five-expert geometry, and the synthetic ZZFeatureMap prose.

> The exact cosine values are cos(72°)≈0.309 and cos(144°)≈−0.809. Therefore, a comment describing all adjacent interference as “+1” is qualitative, not the value computed by the code.

Sources: `quantum_networks.py:168–200; quantum_soul.py:27–30; pineal_gate.py:72–80`

Parameterized gates and deterministic phase injection

LIVE RUNTIME VariationalFractureCircuit, default width 12

Trainable parameterization

PER-LAYER LOCAL ROTATION (4.1)

$$U_{\ell q} = R_z(\phi_{\ell q}) R_y(\theta_{\ell q})$$

The code applies `RY` then `RZ`; the rightmost operator therefore acts first in conventional matrix notation.

ROTATION DEFINITIONS (4.2)

$$R_y(\theta) = [\ \cos(\theta/2)\quad -\sin(\theta/2);\ \ \sin(\theta/2)\quad \cos(\theta/2)\ ],\quad R_z(\phi) = \mathrm{diag}(e^{-i\phi/2}, e^{i\phi/2})$$

These standard gate matrices explain the Qiskit operations; they are not separately implemented as handwritten matrices.

Initialization

ALTERNATING LAYER PRIORS (4.3)

$$\theta_{\ell q} \sim \begin{cases} \mathcal{N}(0, 0.1^2) & \ell \text{ even} \\ -|\mathcal{N}(0, 0.1^2)| & \ell \text{ odd} \end{cases},\quad \phi_{\ell q} \sim \mathcal{N}(0, 0.05^2)$$

The negative absolute-value prior on odd θ layers introduces alternating sign structure.

State-encoding phases

SINGLE-QUBIT PHASE SCHEDULE (4.4)

$$\alpha_{\ell i} = \frac{(\ell+1)(i+1)\pi}{2N_c} = \frac{(\ell+1)(i+1)\pi}{24}$$

Applied as `RZ(αℓi)`.

PAIR PHASE (4.5)

$$\beta_{\ell ab} = \frac{\alpha_{\ell a}\alpha_{\ell b}}{\pi}$$

Applied as `CRZ(βℓab)` on selected state-encoding pairs.

> The active implementation does **not** instantiate Qiskit's `ZZFeatureMap` and does not apply `RZZ`. The pair encoding is a controlled-RZ proxy. "ZZ feature map" appears in comments and training text, but the executable gate is `CRZ`.

Source: `CascadeProjects/windsurf-project/quantum_networks.py:432–545`

Open-system proxy and circuit composition

DETERMINISTIC DRIVE ANGLE (4.6)

$$\gamma_{\ell i} = \frac{(\ell + 1)(i + 1)}{32}\pi$$

Applied as a local `RZ(γℓi)` on the open-system module targets.

PROXY OPERATION	ANGLE	INTERPRETATION IN CODE
`CRZ` on module pairs	$\pi/(\ell+3)$	Layer-dependent damping/phase coupling proxy
`CRX` on entropy-routing pairs	$\pi/7$	Fixed entropy-routing mixing proxy
`CRZ` on readout pairs	$\pi/9$	Fixed measurement/readout coupling proxy

CONCEPTUAL LAYER UNITARY (4.7)

$$U_\ell = U_{\text{readout}}\, U_{\text{entropy}}\, U_{\text{open}}\, U_{\text{encode}}\, U_{\text{entangle}} \prod_{q=0}^{11} U_{\ell q}$$

This expression summarizes application order; the code builds the circuit gate-by-gate from selected topology and module pairs.

What “open system” means here

Implemented	Unitary RZ/CRZ/CRX operations whose names and comments model drive, damping, routing, and readout.
Not implemented	No density matrix, noise channel, Kraus operator, stochastic master equation, or numerical Lindblad integration in the active circuit path.
Consequence	“Lindblad-like” is an architectural analogy for the live gates. The actual Lindblad equation appears only in generated training text.

Observed operation inventory

A direct build of the default active 12-qubit circuit produced:

```
cx: 198   rz: 51   ry: 48   crz: 48
measure: 12   crx: 3   barrier: 3
```

Counts depend on the default layer/topology configuration at audit time. They prove circuit width and gate families, not hardware execution.

Source: `quantum_networks.py:500–545`; verified by local circuit construction

Shot probabilities, Shannon entropy, and marginals

LIVE RUNTIME Classical statistics computed from measured Qiskit counts

EMPIRICAL BIT-STRING PROBABILITY (5.1)

$$p_s = \frac{c_s}{N}, \quad N = \sum_s^{\text{observed}} c_s$$

RUNTIME ENTROPY (5.2)

$$H_2(S) = -\sum_{s:\, p_s > 0}^{\text{observed}} p_s \log_2 p_s$$

This is Shannon entropy of the shot-count distribution, measured in bits.

FITNESS DIVERSITY NORMALIZATION (5.3)

$$H_{\max} = \begin{cases} \log_2 K & \text{if K} > 1 \\ 1 & \text{otherwise} \end{cases}, \quad D = \frac{H_2}{\max(H_{\max}, 1)}$$

K is the number of distinct observed bit strings, not 2^{12}.

SINGLE-QUBIT MARGINAL (5.4)

$$P(q_i = 1) = \sum_{s:\, s_i = 1}^{\text{observed}} p_s$$

READOUT ENTROPY NORMALIZATION (5.5)

$$H_{\text{norm}} = \text{clip}\left(\frac{H_2}{\max(\log_2(\max(K, 2)), 1)}, 0, 1\right)$$

Correct terminology: all entropy produced by the live runtime is count-based Shannon entropy. It is not S(ρ)=−Tr(ρ log ρ).

Sources: `quantum_networks.py:619–641, 946–987`

What the learner optimizes—and what it actually selects

LIVE RUNTIME QuantumLearner evolutionary strategy

FEEDBACK ALIGNMENT (6.1)

$$A = \frac{1}{M} \sum_{m=1}^{M} f_m$$

A is the mean of available expert/hub feedback values; absent feedback yields zero alignment.

PATHWAY DEPTH (6.2)

$$d = \frac{\mathrm{popcount}(s^*)}{N_c} = \frac{\text{active bits in dominant string}}{12}$$

REPORTED CIRCUIT FITNESS (6.3)

$$F = 0.40D + 0.35A + 0.25d$$

Candidate generation

GAUSSIAN MUTATION (6.4)

$$\delta \sim \mathcal{N}(0, \eta^2 I), \quad \theta' = \theta + \delta$$

PARAMETER-ENTROPY SELECTION HEURISTIC (6.5)

$$E_\theta(\theta') = -\sum_{k}^{\text{parameters}} |\theta'_k| \ln(|\theta'_k| + 10^{-10})$$

This is not Shannon entropy of a normalized distribution and not quantum entropy. It is a scalar parameter-spread heuristic.

MOMENTUM AND WRAPPED UPDATE (6.6)

$$v_t = m v_{t-1} + (1-m)\delta_{\text{best}}, \quad \theta_t = (\theta_{t-1} + v_t) \bmod 2\pi$$

Implementation caveat. The measured fitness F is computed and recorded, but the perturbation winner is selected using E_θ, not by evaluating F for each candidate. Consequently, the current update rule does not directly optimize the displayed fitness objective.

Sources: quantum_networks.py:619–705

From measured bits to expert bias and sparse memory

Entropy-based routing

SUBSET MARGINAL DISTRIBUTION (7.1)

$$p_{du} = \sum_{s:\,\mathrm{project}(s,Q_d)=u}^{\mathrm{observed}} p_s$$

Q_d is the qubit subset assigned to routing domain d; u is a projected sub-bit-string.

ROUTING ENTROPY AND BIAS (7.2)

$$H_d = -\sum_{u}^{\mathrm{observed}} p_{du}\log_2 p_{du}, \quad b_d = \mathrm{clip}\left(\frac{H_d}{|Q_d|}, 0, 1\right)$$

Temporal encoding from a trace vector

BOUNDED TEMPORAL ANGLE (7.3)

$$\tilde{\theta}_k = \arctan(z_k) + \frac{\pi}{2}$$

The arctangent maps unbounded trace components into a bounded angular interval shifted by π/2.

FIXED LEARNED/TEMPORAL BLEND (7.4)

$$\theta_k^{\mathrm{blend}} = 0.7\theta_k^{\mathrm{learned}} + 0.3\tilde{\theta}_k$$

Sparse reservoir projection

BAND DECAY (7.5)

$$d_b = 1 - 0.35\frac{b}{11}, \quad b \in \{0, \ldots, 11\}$$

RESERVOIR WEIGHT AND PHASE ADDRESS (7.6)

$$w_r = P(q_{\mathrm{source}} = 1)d_b, \quad a_r = [(r+1)s_{\mathrm{module}}] \bmod 144$$

These are classical records derived from core marginals. They do not update quantum amplitudes.

Sources: quantum_networks.py:756–782, 818–861, 990–1015

The executable 7-qubit fracture circuit

LEGACY FALLBACK Used if the 12-qubit orchestrator is unavailable

The fallback has seven measured qubits: five pentagonal pathway qubits plus Weaver and Void. Its angles are fixed by $\varphi_p=2\pi/5$.

LOCAL PREPARATION (8.1)

$$|\psi_q\rangle = R_x\left(\frac{\varphi_p}{2}\right)R_y(q\varphi_p)\,|\,0\rangle$$

Applied to each of q=0,…,6.

CONNECTION	GATE	ANGLE	ROLE
Pentagon edge (i,(i+1) mod 5)	CRX	φ_p	Nearest-neighbor pathway mixing
Pentagon diagonal	CRZ	$2\varphi_p$	Longer-range phase interference
Weaver couplings	CRX	$\varphi_p/2$	Weaver-mediated mixing
Void couplings	CRZ	φ_p	Void-mediated phase coupling

FALLBACK CIRCUIT SUMMARY (8.2)

$$|\psi_{\text{fallback}}\rangle = U_{\text{Void}}\,U_{\text{Weaver}}\,U_{\text{diagonal}}\,U_{\text{edge}}\prod_{q=0}^{6} R_x(\varphi_p/2)R_y(q\varphi_p)\,|\,0^{\otimes 7}\rangle$$

A direct build produced the following gate inventory:

```
ry: 7   rx: 7   crx: 7   crz: 7   measure: 7
```

> This fallback is the source of many 7-qubit equations in `weaver_reversal_dataset.jsonl`. Those examples describe a real executable fallback, but they do not describe the current default 12-qubit path.

Source: `quantum_soul.py:85–105, 116–169`

Classical mixture-of-experts geometry

CLASSICAL SUPPORT No qubit amplitudes are manipulated in these equations

GATE LOGITS (9.1)

$$g = xW + b$$

BLENDED ROUTING WEIGHTS (9.2)

$$w = 0.5\mathrm{softmax}(g) + 0.5w_{\mathrm{fracture}}$$

The top-k surviving entries are renormalized to sum to one.

CIRCULAR WEIGHTED CENTROID (9.3)

$$c_x = {}^{0}\sum_{i}{}^{4} w_i \cos\vartheta_i\,, \quad c_y = {}^{0}\sum_{i}{}^{4} w_i \sin\vartheta_i$$

$$\vartheta_{\mathrm{centroid}} = \mathrm{atan2}(c_y\,, c_x\,)$$

PAIRWISE INTERFERENCE SCORE (9.4)

$$I = \frac{1}{M} \sum_{i<j}^{\text{active pairs}} \cos(\vartheta_i - \vartheta_j) w_i w_j$$

M is the number of active expert pairs used in the average.

INTERFERENCE GAIN (9.5)

$$G = 1 + 0.2I$$

The gain rescales the combined expert output.

> This is “quantum-inspired” only by analogy: it uses circular phases and a cosine interference term, but all values are ordinary real-valued classical tensors.

Source: `pineal_gate.py:72–80, 252–280, 313–405`

Adaptive classical reservoir ODE

CLASSICAL SUPPORT Continuous-time-inspired state update

ADAPTIVE TIME CONSTANT (10.1)

$$\tau(z) = \tau_{\text{base}}\left[0.1 + 1.9\sigma(w_\tau z + b_\tau)\right]$$

STATE DERIVATIVE (10.2)

$$\frac{dx}{dt} = -\frac{x}{\tau(z)} + \tanh(W_{\text{in}} z + W_{\text{rec}} x + b)$$

STABILIZED EULER STEP (10.3)

$$\Delta t_{\text{eff}} = \min(\Delta t, 0.5\tau), \quad x_{t+1} = x_t + \Delta t_{\text{eff}} \frac{dx}{dt}$$

Routing/scoring around the state

COSINE SCORE AFTER NORMALIZATION (10.4)

$$s_i = \hat{z} \cdot \hat{p}_i, \quad r_i = 0.1 + \max(s_i, 0)$$

INPUT/HUB MIXTURE AND STATE-AMPLIFIED SCORE (10.5)

$$z = 0.8 z_{\text{input}} + 0.2 z_{\text{hub}}, \quad s_i' = r_i (1 + 0.3\|x\|)$$

Final expert scores are normalized after amplification.

This ODE resembles a leaky continuous-time recurrent reservoir. It is classical state-space dynamics; no Hamiltonian, amplitudes, measurement, or density operator is present.

Source: `liquid_fracture.py`

Adaptive governor and hybrid dataset policy

Quantum governor

DORMANT Implemented, but no active runtime consumer was found

ADAPTIVE THRESHOLD (11.1)

$$\tau_t = \tau_{\text{base}} + \lambda \frac{N_{\text{calls}}}{\Delta t}, \quad T_{\max} = 600$$

The equation raises the decision threshold as call rate grows. T_{max} is the maximum call-age/interval constant in the module.

> The dashboard names this component, but repository search found no active runtime import establishing that it governs the live quantum circuit.

Generated hybrid-routing examples

GENERATOR-ONLY forge_fracture_dataset.py and generated weaver_hybrid_dataset.jsonl

GENERATOR ENTROPY (12.1)

$$p = \text{softmax}(\text{logits}), \quad H(G(x)) = -\sum_{i}^{\text{experts}} p_i \ln p_i$$

This generator uses natural logarithms, so H is in nats.

CONDITIONAL HYBRID POLICY (12.2)

$$y = \begin{cases} \text{softmax}(W_g x + \varepsilon W_q x) & \text{if H(G(x))} < \tau\text{t — LOCAL} \\ \text{softmax}(W_g x + \varepsilon \mathbb{E}[O]) & \text{if H(G(x))} \geq \tau\text{t — QPU} \end{cases}$$

The same adaptive threshold form appears with τ_{base}=1.5 and λ=0.5 in generated examples. The JSONL contains numeric training rows; no repository training script was found consuming that hybrid dataset.

Sources: weaver_core/quantum_governor.py; forge_fracture_dataset.py

Omega fuel: feature maps, reduced states, and decoherence

TRAINING TEXT Generated 100 times in Weaver Omega Fuel, with a second declared Bedrock copy

SYNTHETIC ZZ FEATURE INTERACTION (13.1)

$$\mathrm{ZZ}(q_i, q_j) = \exp(i\varphi_p x_i x_j), \quad \varphi_p = \frac{2\pi}{5}$$

This is prose-level feature-map notation in the dataset. It is not the matrix definition of a two-qubit gate and is not how the active circuit is implemented.

REDUCED CORE STATE BY PARTIAL TRACE (13.2)

$$\rho_{\mathrm{core}} = \mathrm{Tr}_{12\,\ldots\,155}(\rho)$$

This equation assumes a 156-qubit density operator. That object does not exist in the live implementation.

VON NEUMANN ENTROPY (13.3)

$$S(\rho) = -\mathrm{Tr}(\rho \log_2 \rho) = -\sum_k^{\text{eigenvalues}} \lambda_k \log_2 \lambda_k$$

The first equality is present in the dataset. The eigenvalue expansion is the equivalent mathematical definition included here to clarify the notation.

LINDBLAD MASTER EQUATION (13.4)

$$\frac{\partial \rho}{\partial t} = -i[H, \rho] + \Gamma\left(L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}\right)$$

$[H,\rho]=H\rho-\rho H$ is a commutator; $\{A,B\}=AB+BA$ is an anticommutator.

SYNTHETIC COHERENCE TIME (13.5)

$$T_2 = \frac{1}{\Gamma}$$

The generator chooses Γ and reports the reciprocal as a synthetic T_2 collapse time.

Sources: `forge_omega_fuel.py:55–111; weaver_omega_fuel.jsonl`

State-space memory and repeated dataset claims

LINEAR STATE-SPACE UPDATE (13.6)

$$h_t = Ah_{t-1} + Bx_t$$

In the generated prose, x_t carries the selected pathway collapse into Mamba-style memory.

Other formulas embedded in declared datasets

FORMULA OR RELATION	DATASET CONTEXT	ACCURACY NOTE
$\dim(\mathcal{H}_n) = 2^n$	Qubit-state dimensionality examples	Standard for n ideal two-level qubits.
H_{max} = n bits	Maximum entropy claim for n measured bits	True for a uniform distribution over all 2^n bit strings.
$S(\rho_A) > S(\rho_{AB})$	Entanglement/conditional-entropy scenario	Possible, not universal. For a pure entangled AB state, S(AB)=0 while S(A)>0.
$RY((2k+1)\varphi_p/2)$	Proposed remedy for fixed-angle pathway bias	Dataset recommendation; not the active initialization formula.
$F = 0.4D + 0.35A + 0.25d$	Reversal examples describing the learner	Matches the runtime’s reported fitness formula.
$\eta_{eff} \approx \eta/(1-m)$	Reversal explanation of momentum overshoot	Rule-of-thumb steady-state amplification, not an exact transient law.
$w = 0.7\theta_{learned} + 0.3\theta_{temporal}$	Temporal encoder discussion	Dataset prose discusses 7 components; live default is 12 qubits.

Corpus facts. `weaver_omega_fuel.jsonl` contains 100 generated records using the same five quantum blocks. A second Bedrock Omega file was declared as another training input in `forge_nemotron.py`. The reversal dataset contains executable fallback and learner formulas. `weaver_soul_dataset.jsonl` contains no occurrence of “quantum” and no quantum equation.

Training completion is not fully provable. `forge_nemotron.py` declares Omega, Bedrock Omega, reversal, and Soul inputs, but the expected `weaver_nemotron_30B_lora` output was absent during audit. The existing Soul adapter does not include a dataset provenance manifest.

Sources: forge_nemotron.py:20–24, 77–82; forge_soul.py; weaver_reversal_dataset.jsonl; weaver_soul_dataset.jsonl

Physics equation found in the sampled checkpoint corpus

EVIDENCE ONLY Separate MoE pretraining checkpoint; not the served Weaver 3B model

The audited checkpoint was at step 6 with 2,359,296 tokens recorded. Reconstructing its 2,304 sampled text windows yielded ten windows containing quantum terms. Only one explicit physics equation family was found.

DRIVEN INTERACTING LATTICE HAMILTONIAN

$$H(t) = -J \sum_{j}^{\text{sites}} (c_j c_{j+1}^{\dagger} + \text{h.c.}) - \mu \sum_{j}^{\text{sites}} n_j + U\cos(\Omega t) \sum_{j}^{\text{sites}} n_j n_{j+1} \tag{14.1}$$

J	Nearest-neighbor hopping scale
c_j, $c_j^{\dagger}$	Annihilation and creation operators at site j
h.c.	Hermitian conjugate of the preceding hopping term
μ, n_j	Chemical potential and local number operator
$U\cos(\Omega t)$	Time-periodic nearest-neighbor interaction strength

This Hamiltonian is corpus exposure evidence only. No source path in the active Weaver runtime constructs or integrates H(t), and this checkpoint was not the locally served `CascadeProjects/3b/weaver_v3` artifact.

Model provenance boundary

Served model	Local Qwen2 3B artifact under `CascadeProjects/3b/weaver_v3`.
Available metadata	Model configuration and safetensors format metadata.
Missing metadata	No complete trainer state or dataset manifest enumerating every upstream equation.
Valid conclusion	The repository supports exact claims about declared local datasets, not exhaustive claims about foundation-model pretraining.

15 · TRUTH MATRIX

Which equations Weaver actually computes

EQUATION FAMILY	LIVE?	WHERE IT EXISTS	CORRECT INTERPRETATION
RY/RZ + CNOT/CRZ/CRX circuit	**Yes**	quantum_networks.py	Active 12-qubit Qiskit circuit
Shot probabilities and Shannon entropy	**Yes**	quantum_networks.py	Classical statistics after measurement
Evolutionary mutation/momentum	**Yes**	quantum_networks.py	Classical parameter update
7-qubit φ_p circuit	Fallback	quantum_soul.py	Executable only when active orchestrator is unavailable
Cosine “interference”	Classical	pineal_gate.py	Real-valued expert-weight geometry
Liquid reservoir ODE	Classical	liquid_fracture.py	Leaky recurrent state dynamics
Adaptive quantum governor	**Not wired**	weaver_core/quantum_governor.py	Dormant threshold logic
156-qubit dense state	**No**	Training prose / architecture labels	144 addresses are sparse metadata
Partial trace $\mathrm{Tr}_{12\ldots155}(\rho)$	**No**	Omega dataset	Training-text scenario
Von Neumann entropy	**No**	Omega dataset	Live code uses Shannon entropy
Lindblad master equation	**No**	Omega dataset	Live code uses unitary gate proxies
Driven lattice Hamiltonian H(t)	**No**	Separate pretraining corpus sample	Exposure evidence only

Dashboard / architecture wording versus executable math

Labels seen in Weaver

- “Lindblad open system”
- “Von Neumann entropy”
- “H_{CR} coupling membrane”
- “156-qubit Heron r2”
- “ZZFeatureMap”

What executes locally

- Deterministic unitary gate proxies
- Shannon entropy from shot counts
- Sparse core/reservoir address mappings
- 12 measured qubits
- RZ + CRZ phase encoding

Bottom line. Weaver contains genuine executable quantum-circuit mathematics, but several stronger open-system and 156-qubit claims belong to architecture metadata or training prose rather than the live numerical implementation.

How this reference was checked

Focused mathematical tests

The repository's targeted unit tests were executed in its virtual environment:

```
venv/bin/python tests/weaver_tests.py --tier unit --test quantum_architecture_reference_math
venv/bin/python tests/weaver_tests.py --tier unit --test quantum_variational_circuit
venv/bin/python tests/weaver_tests.py --tier unit --test quantum_learner_fitness
venv/bin/python tests/weaver_tests.py --tier unit --test quantum_interference_bias
```

Result: all four passed.

Primary codebase source map

REPOSITORY PATH	AUDITED CONTENT
`CascadeProjects/windsurf-project/quantum_networks.py`	12-qubit architecture, geometry, variational circuit, state phases, entropy, fitness, routing, temporal injection, reservoir projection
`CascadeProjects/windsurf-project/quantum_soul.py`	Active orchestrator initialization and legacy 7-qubit fallback
`CascadeProjects/windsurf-project/pineal_gate.py`	Pentagonal MoE geometry and cosine interference
`CascadeProjects/windsurf-project/liquid_fracture.py`	Adaptive classical reservoir dynamics
`CascadeProjects/windsurf-project/weaver_core/quantum_governor.py`	Dormant adaptive threshold
`CascadeProjects/windsurf-project/forge_fracture_dataset.py`	Generated hybrid local/QPU decision equations
`CascadeProjects/windsurf-project/forge_omega_fuel.py`	Omega training-text equations
`CascadeProjects/windsurf-project/weaver_omega_fuel.jsonl`	100 generated Omega records
`CascadeProjects/windsurf-project/weaver_reversal_dataset.jsonl`	Fallback, fitness, entropy, and tuning descriptions
`CascadeProjects/windsurf-project/forge_nemotron.py`	Declared Nemotron dataset inputs
`CascadeProjects/windsurf-project/forge_soul.py`	Soul training path

Completeness statement

"All equations" here means all identifiable quantum equations plus the directly coupled mathematical rules that were found in executable Weaver source, declared local training datasets/generators, and the separately sampled checkpoint corpus. Duplicate occurrences are consolidated into one canonical formula and annotated by status.

This document deliberately excludes ordinary application math unrelated to the quantum subsystem, prose that contains no equation, and uninspectable upstream foundation-model training data.

Audit date: 25 July 2026.
Repository root: `/media/ydn/SYPHER_CORE2/weaver v3`
Document: Weaver Quantum Equations & Notation Reference